

RELATIONS...

* Types of Relations:

- ① Void
- ② Universal
- ③ Reflexive
- ④ Symmetric
- ⑤ Transitive
- ⑥ Equivalence
- ⑦ Identity
- ⑧ Anti-Symmetric

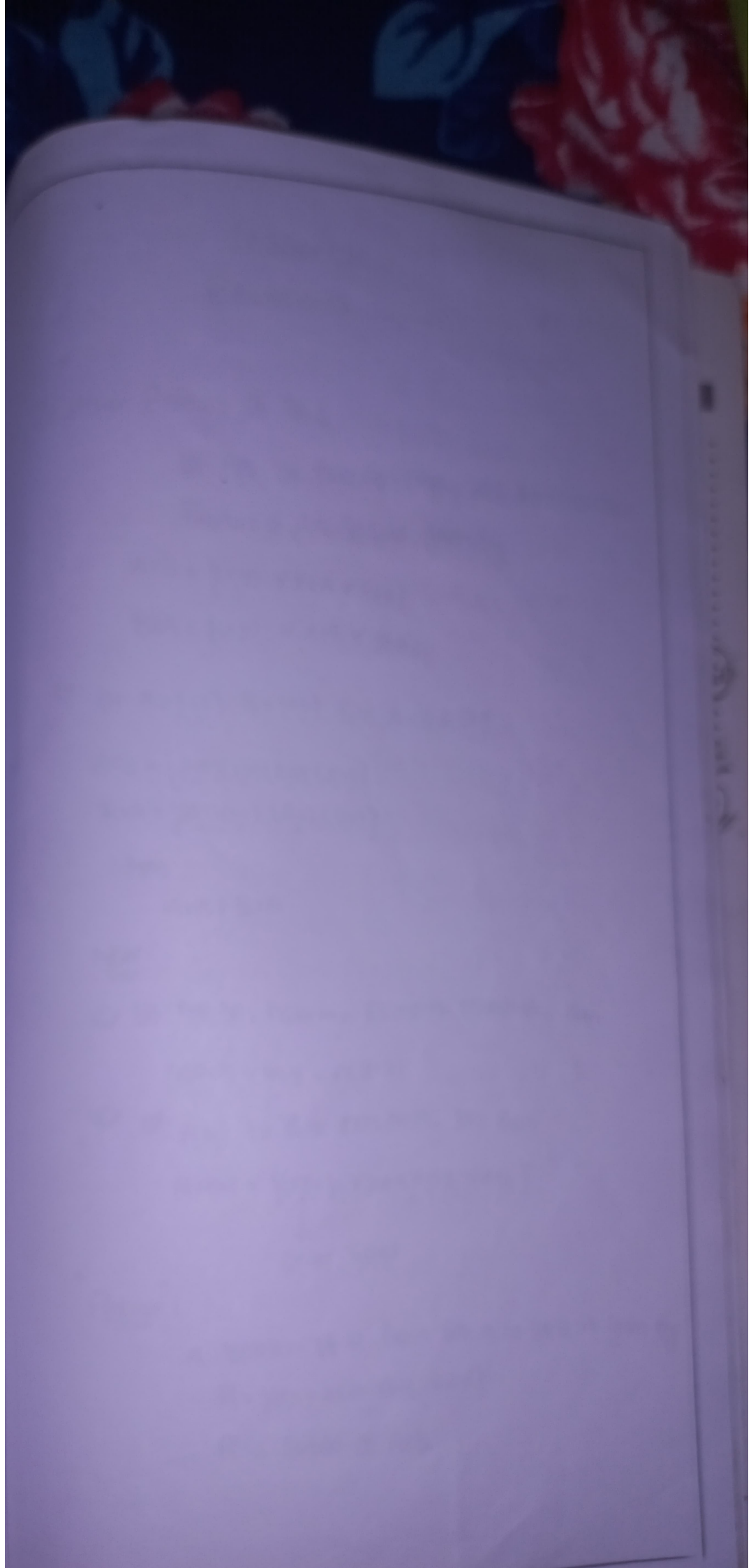
: in Cartesian Product $A \times B \neq B \times A$

: Relation of R from Set A to Set B

$$R = \{(x, y) : x \in A, y \in B, \underline{x R y}\}$$

Means

x is related to y .



LESSON-4

RELATIONS

Cartesian Product of Sets :

Let A, B be two non empty sets then Cartesian Product of A, B Sets is given by.

$$A \times B = \{(x, y) : \forall x \in A, \forall y \in B\}$$

$$B \times A = \{(x, y) : \forall x \in B, \forall y \in A\}$$

ex: Let $A = \{1, 2\}$ $B = \{a, b\}$ find $A \times B$ & $B \times A$.

$$A \times B = \{(1, a), (1, b), (2, a), (2, b)\}$$

$$B \times A = \{(a, 1), (b, 1), (a, 2), (b, 2)\}$$

Where

$$A \times B \neq B \times A.$$

Note:

⇒ Let A, B sets have m, n elements respectively then

$$n(A \times B) = m \cdot n = n(B \times A).$$

⇒ Let A, B, C be three non empty sets then

$$A \times B \times C = \{(x, y, z) : \forall x \in A, \forall y \in B, \forall z \in C\}$$

↓
Order Triplet.

Relation :

A relation of R from Set A to Set B is given by

$$R = \{(x, y) : x \in A, y \in B, x R y\}$$

R is subset of $A \times B$.

Ex: Let $A = \{1, 2, 3\}$ $B = \{4, 5, 6\}$ $R: A \rightarrow B$

- $R = \{(x, y) : x + y > 7\}$

- $R = \{(2, 6), (3, 5), (3, 6)\}$

Q1. $R \rightarrow N \rightarrow N$, $R = \{(x, y) : x^2 + y^2 \leq 10\}$

Q2. $R: \mathbb{Z} \rightarrow \mathbb{Z}$, $R = \{(x, y) : y^2 = 3x \text{ and } y = x\}$

Q3. $A = \{1, 2, 3, 4\}$ $B = \{4, 5, 6, 7\}$ $R: A \rightarrow B$

$$R = \{(x, y) : x = y - 4\}$$

Answers:

- $R = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (3, 1)\}$

- $R = \{(0, 0), (3, 3)\}$

$$y^2 = 3y$$

$$y^2 - 3y = 0$$

$$y(y - 3) = 0 \quad \begin{matrix} y = 0 \\ y = 3 \end{matrix}$$

- $R = \{(1, 5), (2, 6), (3, 7)\}$

Types of Relations:

① Void Relation: A relation R from A to B such that it contains no Order Pair.

Ex:

$$A = \{1, 2\} \quad B = \{3, 4, 5\} \quad R: A \rightarrow B$$

$$R = \{(x, y) : x + y > 10\}$$

Ans: $R = \{\}$

② Universal Relation:

A relation of R from A to B is said to be Universal Relation, if $R = A \times B$.

$$A = \{1, 2\}, B = \{3, 4\} \quad R: A \rightarrow B$$

$$R = \{(x, y) : x + y > 0\}$$

Sol: $R = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$
 $\rightarrow$ Same as $A \times B$

let A be non-empty Set $R: A \rightarrow A$

3. Reflexive relation:

A relation R is said to be reflexive if
 $\forall (a, a) \in R, \forall a \in R$

Ex: let $A = \{1, 2, 3\} \quad R: A \rightarrow A$

1. $R_1 = \{(1, 1), (2, 2)\}$ Not a reflexive relation.
2. $R_2 = \{(1, 1), (2, 2), (3, 3)\}$ ✓
3. $R_3 = \{(1, 1), (2, 2), (3, 3), (1, 2)\}$ ✓

4. Symmetric relation:

A relation R is said to be symmetric
 if R has (a, b) then it must contain $(b, a) \quad \forall (a, b) \in R$

Ex: let $A = \{1, 2, 3\} \quad R: A \rightarrow A$

- ① $R_1 = \{(1, 1), (2, 2)\}$ Symmetric
- ② $R_2 = \{(1, 2), (2, 1)\}$ Symmetric
- ③ $R_3 = \{(1, 3), (3, 1), (1, 2)\}$ Not Symmetric
- ④ $R_4 = \{(1, 2), (2, 3), (3, 2), (2, 1), (1, 1)\}$ Symmetric

5. Transitive relation:

A relation R is said to be Transitive
 if $(a, b), (b, c) \in R$ then $(a, c) \in R$

$$\forall (a, b), (b, c) \in R \quad (or)$$

if $(a, b), (b, c) \in R$ and $(a, c) \notin R$ then it is not transitive.

Ex: Let $A = \{1, 2, 3\}$ $R: A \rightarrow A$

$R_1 = \{(1, 2), (2, 3), (1, 3)\}$ Transitive.

$R_2 = \{(1, 2), (2, 2)\}$ Transitive

$R_3 = \{(1, 3), (3, 2)\}$ Not Transitive!

Equivalence Relation:

A Relation which is Reflexive, Symmetric and Transitive.

$A = \{1, 2, 3\}$ $R: A \rightarrow A$ (R) (S) (T)

- | | | | |
|---|---|---|---|
| 1. $R_1 = \{(1, 1), (2, 2), (3, 3)\}$ | ✓ | ✓ | ✓ |
| 2. $R_2 = \{(1, 1), (2, 2), (3, 3), (3, 2), (2, 3)\}$ | ✓ | ✓ | ✓ |
| 3. $R_3 = \{(1, 1), (2, 2), (3, 3), (1, 3)\}$ | ✓ | ✗ | ✓ |
| 4. $R_4 = \{(1, 1), (2, 2), (1, 3)\}$ | ✗ | ✗ | ✓ |
| 5. $R_5 = \{(1, 1), (2, 3), (3, 2)\}$ | ✗ | ✓ | ✗ |
| 6. $R_6 = \{(1, 1), (3, 3)\}$ | ✗ | ✓ | ✓ |
| 7. $R_7 = \{(1, 3), (2, 2)\}$ | ✗ | ✗ | ⊙ |
| 8. $R_8 = \{(2, 3), (3, 1)\}$ | ✗ | ✗ | ⊗ |
| 9. $R_9 = \{(1, 2), (2, 1), (1, 1)\}$ | ✗ | ✗ | ⊗ |
| 10. $R_{10} = \{(1, 2)\}$ | ✗ | ✗ | ⊙ |
| 11. $R_{11} = \{(3, 3)\}$ | ✗ | ✓ | ✓ |
| 12. $R_{12} = \{\}$ | ✗ | ✗ | ✓ |
| 13. $R_{13} = \{(1, 1), (2, 2), (3, 3), (3, 2), (2, 3), (1, 2)\}$ | ✗ | ✗ | ✗ |